

Visiting Cities

Time limit: 1.00 s

Memory limit: 512 MB

You want to travel from Syrjälä to Lehmälä by plane using a minimum-price route. Which cities will you certainly visit?

Input

The first input line contains two integers n and m : the number of cities and the number of flights. The cities are numbered $1, 2, \dots, n$. City 1 is Syrjälä, and city n is Lehmälä.

After this, there are m lines describing the flights. Each line has three integers a, b , and c : there is a flight from city a to city b with price c . All flights are one-way flights.

You may assume that there is a route from Syrjälä to Lehmälä.

Output

First print an integer k : the number of cities that are certainly in the route. After this, print the k cities sorted in increasing order.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$
- $1 \leq c \leq 10^9$

Example

Input	Output
5 6 1 2 3 1 3 4 2 3 1 2 4 5 3 4 1 4 5 8	4 1 3 4 5